The partial elasticity of demand q with respect to p_1 is defined to be

$$\eta_{qp_1} = \frac{Eq}{Ep_1} = \frac{-p_1}{q}\frac{\partial q}{\partial p_1}$$

The partial elasticity of demand q with respect to p_2 is defined to be

$$\eta_{qp_2} = \frac{Eq}{Ep_2} = \frac{-p_2}{q}\frac{\partial q}{\partial p_2}$$

Example 6.40

Find the marginal productivities of capital (K) and labour (L) if $P = 10L + 0.1L^2 + 5K - 0.3K^2 + 4KL$ when $K = L = 10$.

Solution:

We have $P = 10L + 0.1L^2 + 5K - 0.3K^2 + 4KL$

$$\frac{\partial P}{\partial L} = 10 + 0.2L + 4K$$

$$\frac{\partial P}{\partial K} = 5 - 0.6K + 4L$$

Marginal productivity of labour:

$$\left(\frac{\partial P}{\partial L}\right)_{(10,10)} = 10+2+40 = 52$$

Marginal productivity of capital:

$$\left(\frac{\partial P}{\partial K}\right)_{(10,10)} = 5-6+40 = 39$$

Example 6.41

The production function for a commodity is $P = 10L + 0.1L^2 + 15K - 0.2K^2 + 2KL$ where L is labour and K is Capital.

(i) Calculate the marginal products of two inputs when 10 units of each of labour and Capital are used

(ii) If 10 units of capital are used, what is the upper limit for use of labour which a rational producer will never exceed?

Solution:

(i) Given the production is

$$P = 10L - 0.1L^2 + 15K - 0.2K^2 + 2KL$$

$$\frac{\partial P}{\partial L} = 10 - 0.2L + 2K$$

$$\frac{\partial P}{\partial K} = 15 - 0.4K + 2L$$

When $L = K = 10$ units,

Marginal productivity of labour:

$$\left(\frac{\partial P}{\partial L}\right)_{(10,10)} = 10 - 2+20 = 28$$

Marginal productivity of capital :

$$\left(\frac{\partial P}{\partial K}\right)_{(10,10)} = 15 - 4+20 = 31$$

(ii) Upper limit for use of labour when $K = 10$ is given by $\left(\frac{\partial P}{\partial L}\right) \geq 0$

$$10 - 0.2L+20 \geq 0$$

$$30 \geq 0.2L$$

i.e., $L \leq 150$

Hence the upper limit for the use of labour will be 150 units.

Example 6.42

For the production function, $P = 4L^{\frac{3}{4}}K^{\frac{1}{4}}$ verify Euler's theorem.

Solution:

$P = 4L^{\frac{3}{4}}K^{\frac{1}{4}}$ is a homogeneous function of degree 1.

Marginal productivity of labour is

$$\frac{\partial P}{\partial L} = 4 \times \frac{3}{4} L^{\frac{-1}{4}} K^{\frac{1}{4}} = 3\left(\frac{K}{L}\right)^{\frac{1}{4}}$$

Marginal productivity of capital is

$$\frac{\partial P}{\partial K} = 4L^{\frac{3}{4}} \times \frac{1}{4} K^{\frac{-3}{4}} = \left(\frac{L}{K}\right)^{\frac{3}{4}}$$

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